

# Lee Seou

Industrial Engineering

## CONTACT

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## SKILLS

### Tools

Python   MATLAB

### Numerical Methods

LU Factorization  
Gauss-Seidel   Runge-Kutta  
FDM   FFT  
Interpolation

### Data Analysis

ANOVA   Statistical Analysis

## Capstone Projects

### Cosmetic Manufacturing Emulsion Simulator

#### PBM / Quality Prediction

Population Balance Model (PBM)-based droplet distribution and quality prediction for emulsion manufacturing.

- Responsible for results analysis and project management
- Analyzed droplet size distribution data to evaluate emulsion quality and predict manufacturing outcomes

### Drone Battery Replacement Station Optimization

#### M/M/c Queueing / MATLAB

Capstone project applying M/M/c queueing theory to optimize drone battery replacement station design.

- Built the MATLAB model for the queueing system
- Analyzed system performance under varying station and server configurations

### Autonomous Vehicle TOR Alert Design

#### Human Factors / Team 차오름

Research project studying driver response to Takeover Request (TOR) alerts in autonomous vehicles.

- Designed and ran experiments comparing visual and auditory alert conditions using the Maze platform
- Conducted ANOVA analysis on driver response data

## Education

### B.S. in Industrial Engineering — Sungkyunkwan University

2025.03 ~ Present